PROGRAM 3 – WATER JUG PROBLEM

AIM

To write a Python program to solve the Water Jug problem.

ALGORITHM

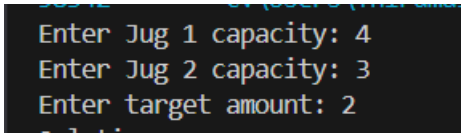
1. Start with both jugs empty.
2. Represent the state using the amount of water in each jug.
3. Generate possible operations.
4. Fill or empty a jug.
5. Pour water from one jug to another.
6. Store visited states.
7. Continue until the target quantity is obtained.

CODE

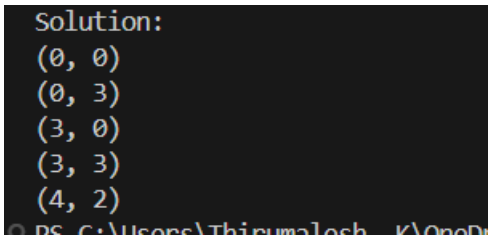
```
from collections import deque
def water_jug(cap1, cap2, target):
    queue = deque([((0, 0), [])])
    visited = set()
    while queue:
        (a, b), path = queue.popleft()
        if (a, b) in visited:
            continue
        visited.add((a, b))
        path = path + [(a, b)]
        if a == target or b == target:
            return path
        states = []
        states.append((cap1, b))
        states.append((a, cap2))
        states.append((0, b))
        states.append((a, 0))
        pour = min(a, cap2 - b)
        states.append((a - pour, b + pour))
        pour = min(b, cap1 - a)
```

```
        states.append((a + pour, b - pour))
        for state in states:
            if state not in visited:
                queue.append((state, path))
    return None
solution = water_jug(4, 3, 2)
print("Solution:")
for state in solution:
    print(state)
```

INPUT:



OUTPUT:



RESULT

Thus, the Water Jug problem was successfully solved using Breadth First Search.